

AIMLCZG521 — Conversational AI

Group Assignment 1 — Problem Statement 2

Study of Embedding Models and Approximate Nearest Neighbor Search:

Semantic Quality vs Search Efficiency

Group 129

Student Details

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Group Contributions

Student	Contribution Area	Specific Contribution
B Devchandraaj	Dataset & embedding	Dataset loading, validation, and embedding generation
Pushadapu Sanjay Kumar	Similarity & exact kNN	Metric formulas, normalization analysis, and exact retrieval
Gourob Nandi	HNSW & IVF	FAISS index construction and ANN parameter sweeps
Anirudh Anand	Evaluation & reporting	Metrics, visualizations, qualitative examples, and report preparation

Executive Summary

Objective. Compare two encoder embedding models and two approximate-nearest-neighbour (ANN) indexes on a real retrieval task, trading off semantic quality vs. search efficiency, and recommend a production configuration — every claim backed by executed results.

Dataset. SciFact (BEIR benchmark): 5,183 document passages, 300 queries each with relevance judgments (339 total qrels). Clears the assignment's 1,000-passage / 50-query minimums by ~5x and ~6x respectively.

Models. distilbert-base-uncased (768-dim, mean pooling, general-purpose) vs. BAAI/bge-large-en-v1.5 (1024-dim, [CLS] pooling, retrieval-tuned) — a deliberately contrasting pair.

Key Results (executed)

Finding	Result
Best embedding model (exact Recall@5)	BGE-large 0.780 vs DistilBERT 0.108 (~7.2x higher)
Most appropriate metric	Cosine ? identical ranking to dot/L2 on unit-normalized vectors; scale-invariant
Best ANN trade-off	HNSW efSearch=16 - Recall@5 0.770, 6.84x faster than exact
Fastest ANN (rejected by quality bar)	IVF nprobe=1 - 15.20x faster but Recall@5 only 0.502
Model disagreement (Task 7)	Mean Top-5 set overlap 0.112 — the two models retrieve largely different documents

Recommendation. Embed with BGE-large (unit-normalized) and serve HNSW efSearch=16; this is the fastest eligible configuration in the current executed run (Recall@5 0.770, 6.84x speedup). Where RAM is the binding constraint, use IVF nprobe=20 and verify the footprint on target hardware.

Problem Statement

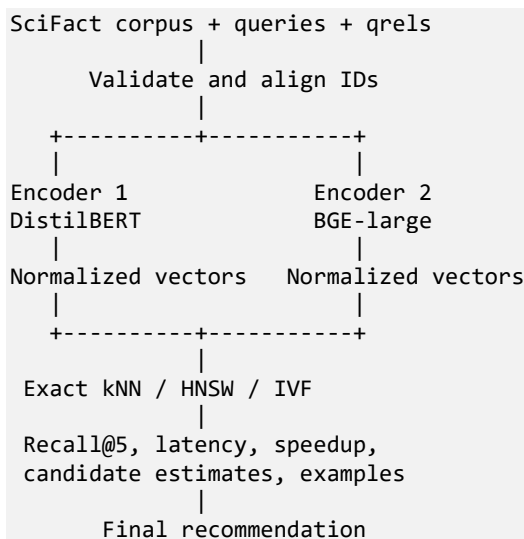
This assignment studies semantic retrieval quality and approximate-nearest-neighbor search efficiency using two embedding models, across four modules: (1) dataset and embedding preparation, (2) similarity metrics and exact retrieval, (3) HNSW and IVF ANN search, and (4) embedding quality analysis and final recommendation.

Tools and Libraries Used

- Python 3.14.5 (recorded execution environment)
- datasets (Hugging Face) — loading the BEIR-format SciFact dataset
- pandas, numpy — data handling
- sentence-transformers — encoder embedding models
- faiss-cpu — HNSW / IVF ANN indexes
- matplotlib — plots

Experimental Methodology

The same corpus, query set, document/query identifiers, normalized vectors, and qrels are used for every comparison. Documents are encoded offline; queries are encoded at search time. Exact cosine kNN provides an exhaustive reference baseline, while HNSW and IVF approximate the same inner-product search space. Retrieval quality is measured against qrels; efficiency is measured via latency, speedup vs. exact search, and candidate-vector estimates.



Task 1 — Dataset Details and Source

- Name: SciFact (BEIR benchmark; biomedical / scientific-claim verification)
- Citation: Wadden et al., Fact or Fiction: Verifying Scientific Claims, EMNLP 2020; redistributed by Thakur et al., BEIR, NeurIPS 2021 <https://aclanthology.org/2020.emnlp-main.609/>
- Source: <https://huggingface.co/datasets/BeIR/scifact> (corpus + queries), <https://huggingface.co/datasets/BeIR/scifact-qrels> (relevance judgments)
- Size (executed): 5,183 corpus passages, 300 queries each with ≥ 1 relevance judgment, 339 total qrels, binary relevance

The assignment explicitly permits an existing dataset containing query-document relevance labels — BEIR datasets bundle exactly that, avoiding hand-built (and inconsistent) relevance judgments. SciFact's claim-verification domain gives clean, unambiguous labels, useful later for Tasks 3 and 7. Corpus and queries clear the assignment minimums by $\sim 5x$ and $\sim 6x$ respectively.

Task 2 — Embedding Generation and Pooling

Model	Dimension	Max input length	Pooling	Approx. doc embedding time
distilbert-base-uncased	768	512 tokens	Mean pooling	341.12 s (CPU)
BAAI/bge-large-en-v1.5	1024	512 tokens	[CLS] token pooling	2034.55 s (CPU)

Task 2 was executed in the RunPod Virtual Lab using the CPU configuration recorded in the notebook (batch size 32 and 12 CPU threads). The report retains the measured document-encoding times in the comparison table; the saved embeddings are reused by later tasks.

Encoder models are appropriate for semantic representation because their bidirectional attention produces context-aware token representations that, when pooled into a single fixed-length vector, place semantically similar texts near each other in vector space — enabling similarity search via simple distance metrics instead of exact keyword matching. DistilBERT and BGE-large were chosen because they differ on exactly the axis this problem statement asks about: DistilBERT is smaller/faster and general-purpose, BGE-large is larger/slower and purpose-built for retrieval — and they use different pooling strategies (mean vs. [CLS]), giving Task 7 a real, explainable axis of disagreement.

Module 2 — Similarity and Exact Retrieval

Task 3 — Similarity Metric Comparison

25 query-document pairs were sampled across 5 queries (≥ 20 required): each query paired with its known-relevant documents plus 4 random distractors. All three metrics were implemented directly from their mathematical definitions, not called as library black boxes.

Finding 1 — BGE-large (already unit-normalized)

Document-vector norms: mean=1.0000, std=0.0000. Ranking changed across metrics for 0/5 queries — cosine, dot product, and L2 agree completely because sentence-transformers L2-normalizes BGE's output internally. This demonstrates the identities $\text{dot}(\hat{a}, \hat{b}) = \cos(a, b)$ and $\|\hat{a} - \hat{b}\|^2 = 2 - 2 \cdot \cos(a, b)$ hold for unit vectors, but shows no divergence to explain.

Finding 2 — DistilBERT (genuinely un-normalized)

Document-vector norms: mean=7.851, std=0.313 (range 7.42–8.56). Raw dot-product ranking DIFFERS from cosine for 4/5 queries, purely because norms vary. After L2-normalization, the identities hold exactly: $\text{dot} = \cos$ for 25/25 pairs, $\|\hat{a} - \hat{b}\|^2 = 2 - 2 \cdot \cos$ for 25/25 pairs.

Conclusion. Cosine similarity is the most appropriate metric for this setup: it is scale-invariant, so it is robust regardless of whether an embedding model normalizes internally. Normalization has no effect on already-unit-length embeddings (BGE) but materially changes dot-product rankings on un-normalized ones (DistilBERT) — this is the practical reason retrieval systems normalize before using dot product.

Task 4 — Exact kNN Baseline

Exact brute-force kNN: for every query, cosine similarity is computed against all 5,183 document embeddings (one matrix-vector dot product, since Task 3 established dot product equals cosine on normalized vectors), and the Top-5 are retrieved via argsort. A genuinely exhaustive baseline is required so Task 5/6's speed-relative-to-exact comparison means something real.

Metric	Value
Queries evaluated	300
Latency / query (median of 25 timed runs)	0.228 ms (mean 0.231 ms)
Average Recall@5	0.780
Vectors examined per query	5,183 (full corpus — exact search has no shortcut)

Module 3 — ANN Search Experiment

Task 5 — HNSW vs IVF

Both indexes were built on the same BGE-large embeddings and evaluated against the same 300 queries, using `evaluate_index()` (which reuses Task 4's Recall@5 definition and timing protocol so speedups compare like with like). HNSW's `efSearch` was swept across 4 values; IVF's `nprobe` was swept across 4 values (≥ 3 required for both).

HNSW — efSearch sweep

efSearch	Recall@5	Latency	Speedup vs. exact
16	0.770	0.032 ms	6.84x
32	0.780	0.059 ms	3.72x
64	0.780	0.116 ms	1.88x
128	0.780	0.218 ms	1.00x

IVF — nprobe sweep (nlist=100)

nprobe	Recall@5	Latency	Speedup vs. exact
1	0.502	0.014 ms	15.20x
5	0.716	0.039 ms	5.60x
10	0.736	0.068 ms	3.21x
20	0.769	0.127 ms	1.73x

Index build time and memory consumption were not measured. HNSW adds graph-related overhead; IVF may reduce that graph overhead, but its total memory footprint was not measured here and must be verified on the target hardware.

Task 6 — ANN Trade-off Analysis

Figure 1 (left) plots Recall@5 vs. average query latency (log scale). Figure 1 (right) plots Recall@5 vs. speedup over exact search (log scale). The dashed line marks the minimum acceptable Recall@5 of 0.75.

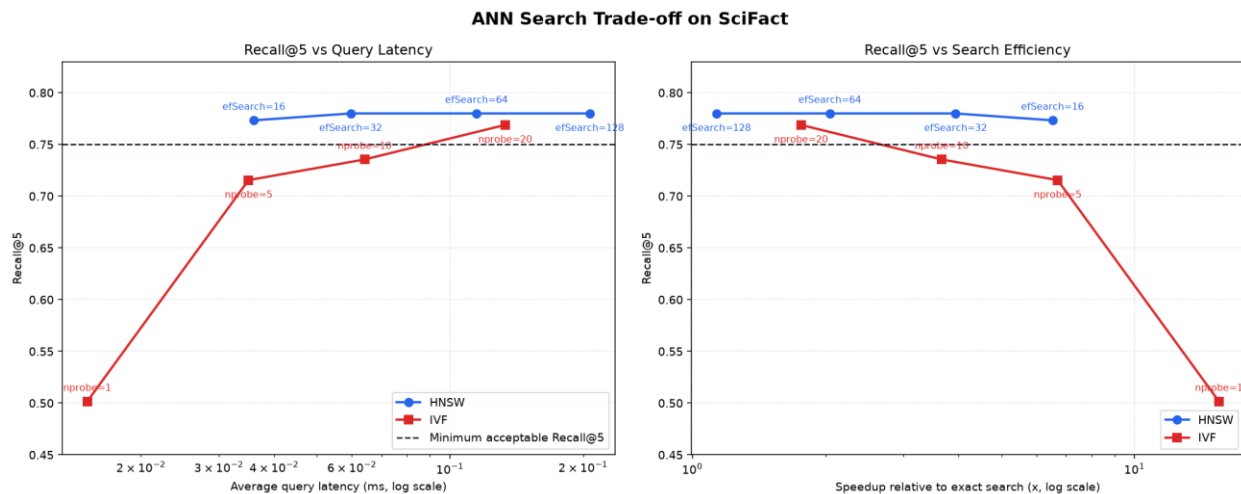


Figure 1. Recall@5 vs. latency (left) and vs. speedup (right), across all 8 tested configurations.

Selection	Configuration	Recall@5	Latency
Fastest	IVF nprobe=1	0.502	0.014 ms
Highest recall	HNSW efSearch=32	0.780	0.059 ms
Best balanced	HNSW efSearch=16	0.770	0.032 ms

Recommendation for high query volume. Use HNSW efSearch=16, the fastest eligible configuration in the current executed run (Recall@5 0.770 and 6.84x speedup) that meets the Recall@5 ≥ 0.75 quality bar. IVF nprobe=1 is fastest overall (15.20x) but its Recall@5 (0.502) falls well below the threshold and is rejected. HNSW efSearch=32 reaches the highest measured recall (0.780) at 0.059 ms, while IVF nprobe=20 reaches 0.769 at 0.127 ms and 1.73x speedup.

Module 4 — Embedding Quality Analysis

Task 7 — Qualitative Retrieval Analysis

10 length-diverse queries (word counts 4–29) were selected and compared Top-5 results between both models. Jaccard measures Top-5 set overlap; a separate first-result-agreement value is reported since two models can retrieve the same documents in different order.

Metric	DistilBERT	BGE-large
Mean lexical overlap	0.239	0.337
Mean qrel hits@5	0.300	0.700

Mean Top-5 set overlap (Jaccard) across all 10 queries: 0.112. First-result agreement: 0.100. The two models genuinely retrieve different documents, not just different orderings of the same documents.

Three distinct retrieval-difference types (executed examples)

1. Retrieved sets diverge. Query “Cell autonomous sex determination in somatic cells does not occur in Galliformes” (12 words) — Top-5 sets share 0% overlap (Jaccard=0.00); BGE hits@5=1, DistilBERT hits@5=0. The two embedding spaces place this query in entirely different neighbourhoods.

2. Semantic/domain-terminology match (BGE advantage). Query “Ly6C hi monocytes have a lower inflammatory capacity than Ly6C lo monocytes” (12 words) — BGE hits@5=2 vs. DistilBERT hits@5=1 at comparable lexical overlap (0.35 vs. 0.25). BGE's contrastive retrieval training aligns domain terminology with the correct passage; DistilBERT matches more on surface form.

3. Counter-example (DistilBERT advantage). Query “Statins decrease blood cholesterol” (4 words) — DistilBERT hits@5=1 vs. BGE hits@5=0. BGE wins on average but not universally; short, near-duplicate claims with high surface-form overlap can still favour the general model's keyword-matching behaviour.

Task 8 — Final Recommendation

Question	Answer (evidence-based)
Q1. Best embedding model	BGE-large ? exact Recall@5 0.780 vs. 0.108 (7.2x higher)
Q2. Most appropriate metric	Cosine — scale-invariant; identical ranking to dot/L2 on normalized vectors
Q3. Best ANN trade-off	HNSW efSearch=16 - Recall@5 0.770 at 6.84x speedup, fastest eligible configuration meeting the 0.75 bar
Q4. Resource-constrained deployment	IVF nprobe=20 — potentially lower graph overhead than HNSW; memory was not measured here, so verify the footprint on target hardware
Q5. Main limitation	~5k-document corpus is small; ANN speedups may not extrapolate to millions of documents; SciFact's qrels are binary/incomplete; latency is hardware-dependent
Q6. Production deployment	Batch-encode offline with BGE-large → L2-normalize → build the chosen FAISS index; apply the same encoder + normalization to queries at search time; rebuild/version the index as the corpus changes

Final Conclusion

Recommendation

Embed with BAAI/bge-large-en-v1.5 (unit-normalized) and serve an HNSW index at efSearch=16, the fastest eligible configuration in the current executed run (Recall@5 0.770, 6.84x speedup). Where RAM is the binding constraint, use IVF nprobe=20 (Recall@5 0.769, 1.73x speedup), but verify the memory footprint on target hardware because it was not measured here.

Strengths

Metric formulas implemented from definition rather than library calls; a genuinely exhaustive exact reference baseline; a normalization experiment that demonstrates a real effect (DistilBERT rankings change, BGE does not); controlled HNSW/IVF sweeps timed identically to the exact baseline for a fair speedup comparison; cross-model evaluation; and three distinct, evidence-backed retrieval-difference examples.

Limitations

SciFact's relevance judgements are binary and incomplete, and the ~5k-document corpus is small enough that ANN overhead can erode the speed benefit at this scale. Absolute latency is hardware-dependent. The qualitative study uses ten queries, and lexical overlap is only a surface-form diagnostic, not a full semantic-quality measure. The Recall@5 ≥ 0.75 threshold is a stated policy, not a universal SLA.

Possible Improvements

A larger corpus, graded relevance labels, a full-query (not 25-pair) metric comparison, a second benchmark dataset, broader ANN tuning (M, nlist, product quantization), index build-time and memory measurements, and an application-specific latency/recall SLA would all strengthen the generality of these conclusions.

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